Translatability FFGFT $\leftrightarrow$ Hilbert-space quantum mechanics

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Preamble

This document formulates the **complete translatability** between the FFGFT mode formalism and standard Hilbert-space quantum mechanics. The translation is not an approximation or specialisation, but a bijective structural statement: every QM calculation can be mapped into FFGFT language, and conversely every FFGFT statement about quantum systems is reproducible in Hilbert-space language.

This is methodologically important: it means that FFGFT does not *replace* quantum mechanics, but *extends* it. QM is a subset of FFGFT (see Doc. 202 §22). Practitioners familiar with Hilbert-space language can use FFGFT without changing their tools – the bridge is the translation table below.

Notation

Geometric spaces

Symbol	Meaning
$\mathbb{R}$	real numbers, linear axis
$\mathbb{C}$	complex numbers
$\mathbb{C}^2$	two-dimensional complex vector space (standard qubit Hilbert space)
$\mathbb{Z}$	integers (for discrete windings)
S^1	circle, one compact loop
S^2	2-sphere, Bloch sphere of standard QM
$T^n = (S^1)^n$	n-torus, n compact loops
T^3	3-torus, three spatial windings (FFGFT mode space)
$T^4 = T^3 \times S^1$	4-torus, full FFGFT geometry incl. time winding
$\mathbb{R} \times S^1$	cylinder, limit of 2-torus with large major radius
$L^2(M)$	space of square-integrable functions on the manifold M
$\mathscr{H}$	Hilbert space (general)

FFGFT geometry parameters

Symbol	Meaning
$\xi_0 = 4/30000$	dimensionless FFGFT geometry parameter
ξ	alternative notation for ξ_0
$D_f = 2.999867$	fractal spacetime dimension
K_{frak}	fractal correction, $K_{\text{frak}} = 1 - 100\xi_0 \approx 0.9867$
$L_0 = \xi_0 \cdot L_P$	minimal FFGFT length scale (sub-Planck)
$L_\xi \approx 100\,\mu$m	emergent vacuum length scale
v	Higgs vacuum expectation value, ≈ 246 GeV
E_0	FFGFT energy scale
m_h	Higgs mass, ≈ 125 GeV
λ_h	Higgs self-coupling
E_ξ	FFGFT energy scale for low-energy limit

Qubit coordinates and states

Symbol	Meaning
$\lvert\psi\rangle$	quantum state (Hilbert-space vector)
$\lvert 0\rangle, \lvert 1\rangle$	computational basis states
α, β	complex amplitudes, $\lvert\alpha\rvert^2 + \lvert\beta\rvert^2 = 1$
z	FFGFT projection on computational axis, $z \in [-1, +1]$
r	FFGFT radial amplitude, $r \in [0, 1]$, $z^2 + r^2 = 1$
θ	FFGFT phase, $\theta \in [0, 2\pi)$
$\varphi_{n,l,j}$	FFGFT mode function on T^4
(n, l, j)	mode quantum numbers
(k_x, k_y, k_z, k_t)	wave vector components on T^4

Operators

Symbol	Meaning
$\sigma_x, \sigma_y, \sigma_z$	Pauli matrices
$\mathbb{1}$	identity matrix
U	unitary operator (quantum gate)
H	Hadamard gate (in gate tables); Hamiltonian (in Schrödinger equation)
T	T phase gate
CNOT	controlled-NOT gate (two-qubit)
$R_{\hat{n}}(\phi)$	Bloch-sphere rotation about axis $\hat{n}$ by angle ϕ
$\hat{n} \cdot \vec{\sigma}$	$n_x\sigma_x + n_y\sigma_y + n_z\sigma_z$
∇^2	Laplace operator

Standard Model quantities

Symbol	Meaning
α	fine-structure constant, $\alpha \approx 1/137.036$
m_i	mass of fermion i ($i = e, \mu, \tau, u, d, c, s, t, b$)
r_i	rational prefactor in $m_i = r_i\xi^{p_i}v$
p_i	winding exponent in $m_i = r_i\xi^{p_i}v$
$\hbar$	reduced Planck constant
c	speed of light

Other symbols

Symbol	Meaning
$\otimes$	tensor product
$\bigotimes_i$	tensor product over index i
$\bigoplus_n$	direct sum over index n
$\lvert x\rangle$	basis state labelled x
$\langle\psi\vert\phi\rangle$	scalar product
$\lvert x\rvert$	absolute value of x
$\arg(x)$	argument (phase) of complex number x
δ_{ij}	Kronecker delta
ϵ_{ijk}	Levi-Civita tensor

Where we are, geometrically

Before the formal translation tables, a brief narrative picture of the underlying geometry. This matters because we are about to work with *cylindrical coordinates* – and the cylinder is not the full FFGFT geometry, but a convenient reduction.

The full FFGFT geometry is four-dimensional. The fundamental space on which all FFGFT statements live is the 4-torus $T^4 = T^3 \times S^1$ – three spatial winding directions and one temporal winding (Doc. 202 §13, Doc. 210). The wave function of a mode $\varphi_{n,l,j}$ is a function on this 4D torus. From this geometry follow all masses, the fine-structure constant, cosmology – everything that distinguishes FFGFT from pure quantum mechanics.

For a single qubit, less suffices. When we want to describe just a single qubit – a two-level system at fixed time, without mass dynamics – most of these winding directions are not active. Two rotational degrees of freedom remain: one encoding "where between $|0\rangle$ and $|1\rangle$ we are" (this becomes our z), and one for the phase (this becomes our θ). The natural space for these two rotational degrees of freedom is a 2-torus $T^2 = S^1 \times S^1$.

From 2-torus to cylinder. If we now assume that one major radius R of the 2-torus is very large compared to the tube radius r, then the local geometry is no longer curved – it looks like a cylinder. The cylinder $\mathbb{R} \times S^1$ is thus the limit $R \to \infty$ of the 2-torus. Topologically one loop is "cut open": from two circles we get a line plus a circle. In practice this means: in cylindrical coordinates (z, r, θ), z is a *linear* axis (from -1 to $+1$), no longer cyclic.

From cylinder to Bloch sphere. If we contract the cylinder top and bottom to a point – the two "poles" – we obtain a sphere S^2. This is the famous **Bloch sphere** of standard quantum mechanics. It is the second, further reduction: cylinder with merged ends.

Hierarchy of reductions.

Geometry	Dim.	Use
4-torus T^4	4	full FFGFT (masses, α, cosmology)
3-torus T^3	3	mode space at fixed time
2-torus T^2	2	single qubit, without reduction
Cylinder $\mathbb{R} \times S^1$	2	limit $R \to \infty$, local approximation
Bloch sphere S^2	2	poles contracted, standard QM

What the reductions cost. Each step of reduction loses something. From the 4-torus to the 3-torus one loses the time winding – and with it all mass statements (Doc. 210). From the 3-torus to the 2-torus one loses a spatial winding – and with it the connection to the third lepton generation and many spin structures. From the 2-torus to the cylinder one loses *a compact loop* – this is the weakest reduction, and exactly here sits the **fractal correction factor** $K_{\text{frak}} \approx 0.9867$ that appears in the bit-flip rotations of Doc. 175. K_{frak} is the "memory" of the cylinder of its toroidal origin.

What we translate here. The translation tables below apply to the *qubit sector*, i.e. the lower three rows of the hierarchy – 2-torus, cylinder, Bloch sphere. Standard quantum mechanics works on the Bloch sphere; FFGFT works on the cylinder (or the 2-torus, if one takes K_{frak} seriously). The translation between these levels is lossless, provided one carries the small geometric corrections (K_{frak}, ΔCHSH) along.

The full FFGFT geometry – the 4-torus – lies one step higher and provides what Hilbert-space QM cannot provide: the values of the input parameters (masses, α, ξ). This is discussed in §10 below.

.1 States: (z, r, θ) vs. (α, β)

Hilbert-space representation

In standard QM a qubit is a vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1, \qquad \alpha, \beta \in \mathbb{C}. \tag{1}$$

With global phase freedom this is a point on the Bloch sphere (Hopf fibration $S^3 \to S^2$).

FFGFT representation

In FFGFT the same qubit is a point (z, r, θ) on the cylinder/torus:

$$z \in [-1, 1], \quad r \in [0, 1], \quad \theta \in [0, 2\pi), \qquad z^2 + r^2 = 1. \tag{2}$$

z is the projection onto the computational basis axis, r the radial superposition amplitude, θ the phase (Doc. 147 §1.2).

Bijective translation

The two representations are linked by:

$$\boxed{\alpha = \sqrt{\tfrac{1+z}{2}} \cdot e^{i\theta/2}, \qquad \beta = \sqrt{\tfrac{1-z}{2}} \cdot e^{-i\theta/2}.} \tag{3}$$

Conversely:

$$z = |\alpha|^2 - |\beta|^2, \qquad r = 2|\alpha\beta|, \qquad \theta = \arg(\alpha) - \arg(\beta). \tag{4}$$

Consistency checks:

- $|\alpha|^2 + |\beta|^2 = \frac{1+z}{2} + \frac{1-z}{2} = 1$. ✓
- $z^2 + r^2 = (|\alpha|^2 - |\beta|^2)^2 + 4|\alpha|^2|\beta|^2 = 1$. ✓
- $|0\rangle$ corresponds to $(z, r, \theta) = (1, 0, *)$. ✓
- $|1\rangle$ corresponds to $(z, r, \theta) = (-1, 0, *)$. ✓

Conceptual difference: in QM, $|\alpha|^2$ is a *probability* (Born rule). In FFGFT, $r^2 = 1 - z^2$ is an *energy density* of the radial configuration. The two quantities are *numerically equal* at large statistics (Doc. 147), but conceptually different: FFGFT is a deterministic geometric model whose statistical predictions coincide with the Born rule.

.2 Operators: Pauli matrices vs. rotations

Hilbert-space operators

The three Pauli matrices generate all single-qubit operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{5}$$

Algebra: $\sigma_i \sigma_j = \delta_{ij}\mathbb{1} + i\epsilon_{ijk}\sigma_k$.

FFGFT rotations

On the Bloch cylinder, the three Pauli matrices correspond to three geometric rotations (Doc. 175 §3):

- σ_z – rotation about the cylinder axis (phase rotation $\theta \to \theta + \pi$).
- σ_x – 2D rotation in the (z, r) plane by angle π (bit flip).
- σ_y – orthogonal 2D rotation, combining σ_x and phase rotation.

Translation table

Operator	Hilbert-space matrix	FFGFT geometry
σ_z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Phase rotation $\theta \to \theta + \pi$
σ_x	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	(z, r)-rotation by πK_{frak}
σ_y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	Rotation with phase tilt
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	Swap $z \leftrightarrow r$, $\theta \to \theta + \pi/2$
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	Phase rotation $\theta \to \theta + \pi/4$

The only FFGFT-specific deviation is the factor $K_{\text{frak}} \approx 0.9867$ in the σ_x rotation (Doc. 175). This deviation follows from the fractal spacetime dimension $D_f = 2.999867$ and is a *testable prediction*: all π-gates deviate systematically by $\Delta\alpha \approx 0.013\,\%$ from the ideal value. In Hilbert-space formalism this would be an ad hoc correction modeled as hardware noise; in FFGFT it follows from the geometry.

.3 Quantum gates: complete translatability

Single-qubit gates

Every unitary single-qubit gate $U \in U(2)$ can be represented as a Bloch-sphere rotation:

$$U = e^{i\alpha} R_{\hat{n}}(\phi) = e^{i\alpha}(\cos(\phi/2)\mathbb{1} - i\sin(\phi/2)\,\hat{n}\cdot\vec{\sigma}). \tag{6}$$

In FFGFT this is a rotation of the (z, r, θ) point about axis $\hat{n}$ by angle ϕ. The translation is exact: $U|\psi_{\text{QM}}\rangle$ corresponds to the rotation of the FFGFT cylinder point about $\hat{n}$ by ϕ.

Two-qubit gates

The CNOT gate:

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{7}$$

translates in FFGFT as: *if $z_A = -1$ (control qubit on $|1\rangle$), perform a bit flip on qubit B*. This is geometric conditioning: the rotation of one cylinder point depends on the z value of the other. Topologically this corresponds to a linkage of the two cylinders (entanglement).

Universality

The set $\{H, T, \text{CNOT}\}$ is universal for QM (Solovay-Kitaev theorem). Since each of these gates is representable as a geometric operation in FFGFT, FFGFT is *fully universal for quantum computation*. There is no QM statement that cannot be formulated in FFGFT.

.4 Tensor products and many-qubit spaces

Hilbert-space construction

For n qubits the Hilbert space is $\mathscr{H}_n = \bigotimes_{i=1}^{n} \mathbb{C}^2$ with dimension 2^n. A general state is:

$$|\Psi\rangle = \sum_{x\in\{0,1\}^n} \alpha_x |x\rangle, \qquad \sum_x |\alpha_x|^2 = 1. \tag{8}$$

FFGFT product space

In FFGFT n qubits correspond to n independent mode configurations (z_i, r_i, θ_i) on a shared torus geometry. Entanglement is represented as a *topological linkage*: entangled qubits are nodes of the mode configuration that cannot be reduced to product form (Doc. 175 §4).

Bell states

The maximally entangled Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{9}$$

has in FFGFT a geometric characterisation as a *topologically linked cylinder configuration*, in which the individual (z_i, r_i, θ_i) values are not independent.

CHSH prediction: standard QM gives $|\langle\text{CHSH}\rangle| \leq 2\sqrt{2}$ (Tsirelson bound). FFGFT predicts a small, geometrically motivated deviation:

$$\Delta\text{CHSH} \approx 10^{-5} \qquad \text{(geometric, from } \xi \text{ and } K_{\text{frak}}). \tag{10}$$

This deviation lies below the current hardware noise floor and is an open experimental task for high-precision Bell tests (Doc. 175 §6, Doc. 023).

Terminology note: $\Delta\text{CHSH} \approx 10^{-5}$ is the *elementary* deviation per measurement, derived from $\xi/(2\pi)$. This is distinct from the *cumulative* value at $N = 73$ measurements in Doc. 022 ($\approx 5 \times 10^{-4}$). The two quantities are not directly comparable; 10^{-5} is not reachable at any finite N from the cumulative formula in Doc. 022.

Exponential growth

Both formalisms reproduce the exponential growth dim $= 2^n$. In Hilbert space through tensor product; in FFGFT through n independent mode configurations with topological entanglements. The number of real parameters per state is in both cases $2 \cdot 2^n - 2$.

.5 Measurement

Hilbert-space Born rule

A measurement in the computational basis yields outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$. The state collapses to $|0\rangle$ or $|1\rangle$ respectively.

FFGFT measurement as polar transition

In FFGFT, measurement is *not* a probabilistic collapse, but a *geometric interaction* that drives the (z, r, θ) point to the nearest stable extremum ($z = +1$ or $z = -1$). This is a deterministic motion in mode space – but due to the unknown phase θ and tiny variations in initial conditions, it is *statistically* indistinguishable from Born-rule behaviour.

Numerical equivalence

At large statistics (many repetitions), the FFGFT measurement distribution converges to the Born rule:

$$P_{\text{FFGFT}}(z = +1) = \tfrac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0). \tag{11}$$

The two formalisms yield *identical* statistical predictions.

Conceptual difference: the Bell-test realism debate resolves geometrically in FFGFT, since entanglement is represented as a topological link – not as nonlocal correlation. The experimental predictions are the same.

.6 Unitary evolution: Schrödinger equation

Hilbert-space Schrödinger

The time evolution in QM is:

$$i\hbar\frac{\partial|\psi\rangle}{\partial t} = H|\psi\rangle. \tag{12}$$

FFGFT mode evolution

In FFGFT the time evolution is the deterministic motion of the mode configuration $\varphi_{n,l,j}$ on the torus $T^3{\times}\mathbb{R}$ (Doc. 202 §13). The corresponding equation of motion – the FFGFT mode equation – reduces in the low-energy limit ($E \ll E_\xi$) to the Schrödinger equation:

$$i\hbar\frac{\partial\varphi_{n,l,j}}{\partial t} \xrightarrow{E\ll E_\xi} -\frac{\hbar^2}{2m}\nabla^2\varphi_{n,l,j} + V\varphi_{n,l,j}. \tag{13}$$

At higher energies, ξ corrections appear, which in the Hilbert-space formalism manifest as Hamiltonian modifications.

.7 What is fully translatable

All Hilbert-space statements map into FFGFT:

- States, operators, gates (with or without K_{frak} correction),
- Tensor products, entanglement, Bell states, GHZ states,
- Measurement (statistical predictions identical to Born rule),
- Unitary evolution (Schrödinger in low-energy limit),
- Algorithms (Shor, Grover, VQE, etc.).

All FFGFT QM statements map into Hilbert space:

- The (z, r, θ) configuration corresponds uniquely to the (α, β) vector (bridge (3)),
- Geometric rotations correspond to unitary operators,
- Topological entanglements correspond to entangled Hilbert-space vectors.

Consequence: every computation that can be performed in one formalism can be performed in the other – with identical end result. Practitioners can choose the formalism familiar to them.

.8 What is FFGFT-specific and not representable in pure Hilbert space

The translatability holds for *statements about quantum systems*. It does not hold for *statements about the input values* of quantum mechanics:

- The geometric parameter $\xi_0 = 4/30000$ appears in the Hilbert-space formalism only as an external input. In FFGFT it follows from Higgs matching $\xi_0 = \lambda_h^2 v^2/(16\pi^3 m_h^2)$ (Doc. 006).
- The fermion masses $m_e, m_\mu, m_\tau, m_u, \ldots$ appear in Hilbert space as external Yukawa couplings. In FFGFT they all follow parameter-free from $m_i = r_i \xi_0^{p_i} v$ (Doc. 006).
- The fine-structure constant $\alpha = \xi_0 \cdot E_0^2$ is derivable in FFGFT, an input in Hilbert space.
- The fractal correction factor $K_{\text{frak}} = 1 - 100\xi_0 \approx 0.9867$ appears in FFGFT gates as a geometric deviation. In Hilbert space it would have to be modeled as hardware noise.

This is the structural asymmetry:

- Hilbert space: has all tools for QM calculations, but no explanation for the input values (masses, couplings, α).
- FFGFT: provides both the tools for QM calculations *and* the input values from a single geometry.

In this sense QM is a subset of FFGFT (see Doc. 202 §22): QM is FFGFT, restricted to the question *how systems behave*, without the question *why the constants take these values*.

.9 Consequence: extension, not replacement

FFGFT does not replace quantum mechanics. It extends it to a level where the input values themselves follow from geometry.

For practitioners this means:

- Those who calculate with Hilbert-space tools can continue to do so. The translation (3) and the operator table allow switching to the FFGFT formalism at any time, without loss.
- Those who calculate with the FFGFT mode formulation additionally have access to masses, couplings, and α from the same geometry.
- Both languages are *interoperable*. This is precisely the "translatability without contradiction" from Doc. 202 §22: other frameworks can map FFGFT statements into their language, provided they use the translation table.

Methodological clarification: FFGFT does not claim to be the only correct representation of quantum mechanics. It is *one* representation that, in addition to the standard QM statements, provides the values of the fundamental constants. Those who do not need this added value can ignore FFGFT – the standard QM statements remain valid.

.10 Summary

The translation table FFGFT ↔ Hilbert-space QM is complete:

1. **States:** bijective bridge $\alpha = \sqrt{(1+z)/2}\,e^{i\theta/2}$, $\beta = \sqrt{(1-z)/2}\,e^{-i\theta/2}$.
2. **Operators:** Pauli matrices ↔ Bloch rotations, with FFGFT correction K_{frak} in σ_x.

3. **Gates:** all standard gates (H, T, CNOT, ...) in FFGFT as geometric transformations.
4. **Tensor products:** 2^n growth reproduced, entanglement as topological linkage.
5. **Measurement:** statistically identical to Born rule, conceptually deterministic geometry.
6. **Evolution:** Schrödinger equation as low-energy limit of FFGFT mode evolution.

FFGFT added value beyond Hilbert space:

- parameter-free derivation of α, m_i, ξ from a single geometry,
- geometric explanation of the fractal correction factor K_{frak} in π-gates (testable in high-precision quantum gate experiments),
- geometric resolution of the Bell-test realism debate (entanglement as topology, not as non-locality),
- small, geometrically motivated CHSH deviation $\sim 10^{-5}$ as a testable prediction.

Methodological position: FFGFT confirms quantum mechanics and extends it. It refutes nothing, but adds structural explanations for the input values. The translatability ↔ Hilbert space is a *consistency proof*: any internally consistent theory should be expressible in multiple mathematical languages, and the existence of such translations is evidence of structural soundness.